

Appendix A

A Linux system administrator who doesn't know shell scripting is like batman without BatMobile, with the Bat suite you will still like one, but that is about it. This is a quick and handy introduction to Bash scripting, it is written with the assumption that you have no previous knowledge of bash scripting.

Hello World and Variables

Open your favourite text editor (use vi, emacs, gedit or your own version) and save the following code in a file and name it as you fancy, but with a *.sh extension.

Code

```
#!/bin/bash
# declare a variable by the name of outP
outP="Hello World"
# displaying the variable on a screen
echo outP
# displaying a literal string on screen
echo 'hello digit readers'
```

Explanation

#! stands for **shebang**. It informs the system about the interpreter for which this script is intended to run – in this case **Bash** located in the **/bin/bash** location. # as you must have guessed by now, indicates a comment, and only in the first line as a part of shebang symbol does it have special meaning.

Next comes declaration of the variable, now it may seem natural to Python and like programmers, but people coming from C/C++ backgrounds should take note that neither do you need to declare a variable before using it nor do you need to specify a type. The type of variable is automatically determined by the content and how it is treated.

A variable is referred as **\$Var_Name**, without the quotes, in the print syntax to distinguish it from a immutable string which will be represented by enclosing in quotes. The print syntax is the echo command. Echo is equivalent to **print** command in Python, **cout** in c++ and **printf** in C.

We will end this introduction after briefly mentioning about Array Variables. You can consider the following example which will illustrate declaring and accessing array variables and individual elements inside it.

```
#!/bin/bash
#Array declaration of elements
Digit_Contents=( 'DVD' 'Fast_Track' 'Digit_Magazine' )
```